

GLOBAL CYBERSECURITY THREATS

A Hypothesis-Driven Risk Analysis (2015–2024)

End-to-End Analytics Project Report — Excel · SQL · Power BI

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 Data Limitation: This project uses synthetic data intended for analytical methodology demonstration. Findings should not be interpreted as real-world estimates of global cybersecurity risk.

1. Executive Summary

This project analyzes a global cybersecurity incident dataset (2015–2024) using a hypothesis-driven methodology rather than a purely descriptive approach. Five specific hypotheses about attack severity, financial loss, time trends, industry exposure, and operational response were evaluated using descriptive, comparative, and formal statistical analysis across Excel, SQL, and Power BI — each carried through a consistent chain: Question → Data → Metric → Analysis → Finding → Business Meaning → Decision.

Two of the five hypotheses were rejected in their strong form, one uncovered a mathematical artifact in a metric designed for this project, and the single high-confidence positive finding only held after narrowing the analysis from average risk to tail risk — a distinction that became the project's central insight: average risk and tail risk are not the same signal.

Core Insight: "Average Risk Looks Equal — Tail Risk Does Not." No country, industry, or attack type stands out on average, but USA-associated DDoS and Ransomware incidents are disproportionately concentrated among the highest-loss cases.

2. Business Problem

The organization lacks a consolidated, evidence-based view of its cybersecurity incident history. Existing reporting tends to rely on simple counts and averages ("which attack happens most"), which — as this project demonstrates — can mask where the real financial and operational risk is concentrated.

Business Questions Addressed

- What does the threat landscape look like, and is it worsening over time?
- Which systems, industries, or countries carry disproportionate risk?
- Are current defense mechanisms measurably reducing financial loss?
- Does incident severity actually predict slower operational response?

3. Dataset & Methodology

3.1 Dataset

Source: Global Cybersecurity Threats (2015–2024), Kaggle

Columns: Country, Year, Attack Type, Target Industry, Financial Loss (in Million \$), Number of Affected Users, Attack Source, Security Vulnerability Type, Defense Mechanism Used, Incident Resolution Time (in Hours)

Records: 3,000 incidents

 Important: This dataset is synthetic. Distributions across country, industry, attack type, and year are close to uniform — consistent with randomly generated data rather than real-world incident records. All findings are

framed as patterns observed within this dataset, not as real-world cybersecurity conclusions.

3.2 The Analytical Framework

Rather than executing a fixed checklist (build column → build pivot → build chart), every analysis in this project followed a single reasoning chain:

Question → Data → Metric → Analysis → Finding → Business Meaning → Decision

This discipline is what allowed the project to catch and correct its own mistakes — most notably in Hypothesis 5, where a derived metric (Resolution Efficiency) was found to produce a misleading result and was demoted from primary KPI to a supporting, carefully-caveated metric rather than being reported at face value.

3.3 Four Analytical Lenses

Lens	Business Question	Related Hypotheses
A. Threat Landscape	What does the threat look like, and how does it evolve?	H3
B. Risk Concentration	Who is truly exposed — beyond the average?	H2, H4
C. Defense Effectiveness	Are current defenses measurably reducing loss?	H2 (supporting)
D. Severity & Response	Does severity actually predict response speed?	H1, H5

4. Phase 1 — Excel: Data Cleaning & Exploration

4.1 Data Cleaning

- Removed duplicates and validated missing values
- Standardized text casing and fixed acronym distortion (e.g., "It" → "IT", "Sql Injection" → "SQL Injection")
- Verified numeric fields (Financial Loss, Resolution Time, Affected Users) were valid and non-negative

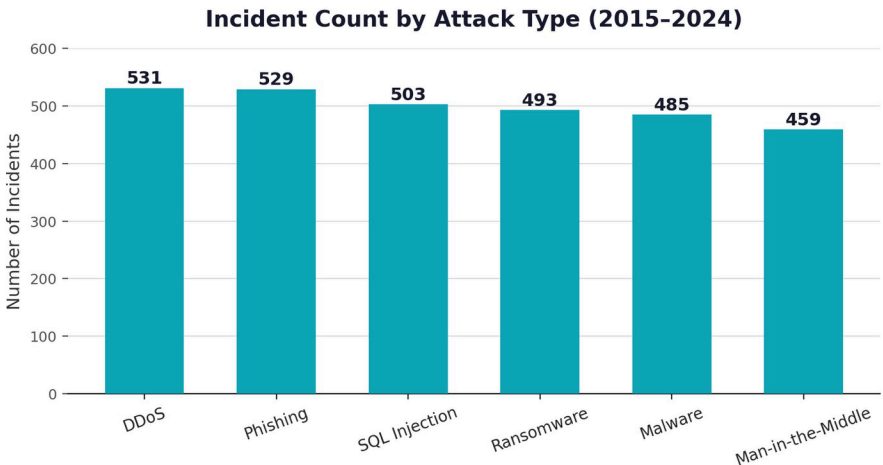
4.2 Calculated Columns — Final Status

Column	Status	Final Role
Severity_Score	Adopted	Composite classification metric — disclosed as partly structural with Financial Loss
Severity_Level	Adopted	Low / Medium / High / Critical classification — basis for Hypothesis 1
Cost_per_User	Adopted	Direct economic metric; no distortion detected
Resolution_Efficiency	Supporting only	Used in SQL analysis; NOT surfaced as a primary Power BI KPI
Decade_Period	Discarded	Yearly analysis (Hypothesis 3) proved more precise; would have hidden the 2018 compositional shift

Principle: no calculated column is treated as final simply because it was planned. Its role in the analysis (primary KPI, supporting metric, or discarded) is decided only after testing it against the data.

4.3 Cross-Sectional Pivot Analysis (8 checks)

Attack Type, Severity, Financial Loss by Country/Industry/Attack Type, Attack × Severity cross-tab, yearly trend, and Defense Mechanism were each examined independently. Across all eight checks, incidents were distributed in a narrow range across categories — no single dimension stood out sharply in isolation. This result itself became the starting point for the deeper, multi-dimensional SQL analysis in Phase 2.



Incident counts are distributed within a narrow band (459–531) across all six attack types — no single type is dominant by volume.

5. Phase 2 — SQL: Hypothesis-Driven Analysis

Instead of re-running Excel's descriptive pivots in SQL, five specific hypotheses were evaluated using CTEs, window functions (RANK, NTILE), and aggregate comparisons — each hypothesis stated, examined, and resolved with an explicit evidence strength rating rather than a formal statistical significance level.

Hypothesis 1. Critical incidents have substantially higher financial losses

Finding: Critical-labeled incidents average \$83.8M in loss vs \$17.1M for Low-labeled incidents.

Why it matters: The relationship is partly structural, because Financial Loss is an input to the Severity Score. It validates the internal construction of the index but does not provide independent evidence that severity predicts financial loss.

Decision: Do not use this finding as standalone justification for operational resource allocation.

Recommendation: Use the score as an internal analytical prioritization tool, subject to validation against real operational outcomes before any operational deployment.

Evidence Strength: Moderate

Hypothesis 2. Certain attack types are responsible for disproportionately high losses

Finding: Rejected at the average level (all attack types fall within a ~5% band of the global average). Confirmed at the tail level: USA × DDoS (12 occurrences) and USA × Ransomware (10 occurrences) appear more often within the top-loss decile than any other combination in the same decile.

Why it matters: This is the strongest and most actionable pattern in the dataset — it shows that average risk and tail risk are not the same signal.

Decision: Flag USA-associated DDoS and Ransomware incidents for closer review, rather than treating USA as average-risk based on overall loss figures.

Recommendation: Within this dataset, these combinations warrant further investigation and targeted monitoring — not immediate resource reallocation, given the synthetic nature of the data.

Evidence Strength: Strong descriptively

Hypothesis 3. Attack frequency changes significantly over time

Finding: Rejected for overall volume (263–319 incidents/year, no trend). Partially supported for composition: SQL Injection was the top-total-loss attack type in 5 of 10 years (2018–2023), despite ranking below the global average in per-incident loss (50.01 vs 50.49) — its yearly dominance is driven by incident-volume concentration, not per-incident severity.

Why it matters: Distinguishes two different kinds of "dominance": total-loss-in-a-year (volume effect) versus severity-per-incident (which SQL Injection does not have).

Decision: Monitor SQL Injection at the year level; do not treat it as the most severe attack type per incident.

Recommendation: Treat this as an observed, recurring pattern — not a statistically validated trend, since year-over-year margins were often narrow.

Evidence Strength: Weak–Moderate

Hypothesis 4. Certain industries are more exposed to specific attack types

Finding: Rejected in its strong form — no industry shows any single attack type exceeding ~22% of its incidents (vs. a ~16.7% uniform baseline). A weak descriptive pattern only: Banking (21.6%) and Retail (21.0%) both peak on Phishing.

Why it matters: No sharp industry specialization exists in this dataset.

Decision: Do not design industry-specific defenses based on this dataset alone.

Recommendation: Treat the Banking/Retail–Phishing pattern as a minor descriptive note, not a validated finding.

Evidence Strength: Weak

Hypothesis 5. More severe / certain-category incidents take longer to resolve

Finding: Direct Resolution Time shows no meaningful difference across Severity, Attack Type, or Defense Mechanism (35–37 hours uniformly). The derived Resolution Efficiency ratio shows a misleading 6.5x spread by Severity (Low=295.5 vs Critical=44.3) — a mathematical artifact caused by a near-zero Severity_Score denominator for the Low band, not a real operational difference.

Why it matters: Demonstrates why relying on a single composite ratio can produce false conclusions. Using two metrics side by side caught this artifact before it could be misreported as a real 6.5x efficiency gap.

Decision: Resolution Time becomes the primary operational KPI in the dashboard. Resolution Efficiency is retained only as a secondary, supporting analytical metric — never surfaced as a headline "Effectiveness" KPI.

Recommendation: Resolution_Efficiency should be used only between categories with reasonably-sized Severity_Score denominators (Attack Type, Defense Mechanism), and must be avoided as a standalone Severity_Level comparison.

Evidence Strength: Strong descriptively (Time) / Not applicable (Efficiency by Severity)

Hypothesis Evaluation Summary

#	Hypothesis	Result	Evidence Strength
1	Critical → higher loss	Confirmed, partially circular	Moderate
2	Attack type → outsized loss	Rejected (avg) / Confirmed (tail)	Strong descriptively
3	Frequency changes over time	Rejected (volume) / Partial (composition)	Weak–Moderate
4	Industry × Attack specialization	Rejected (strong form)	Weak

#	Hypothesis	Result	Evidence Strength
5	Severity → resolution speed	Rejected (Time) / Artifact (Efficiency)	Strong (Time) / N-A

No hypothesis was confirmed at strong evidence in its general, dataset-wide form. The one hypothesis with strong descriptive support (H2) only holds when narrowed from average risk to tail risk — the project's central methodological takeaway. Evidence Strength here reflects the consistency and actionability of the descriptive pattern observed, not a formal statistical significance level (see Limitations).

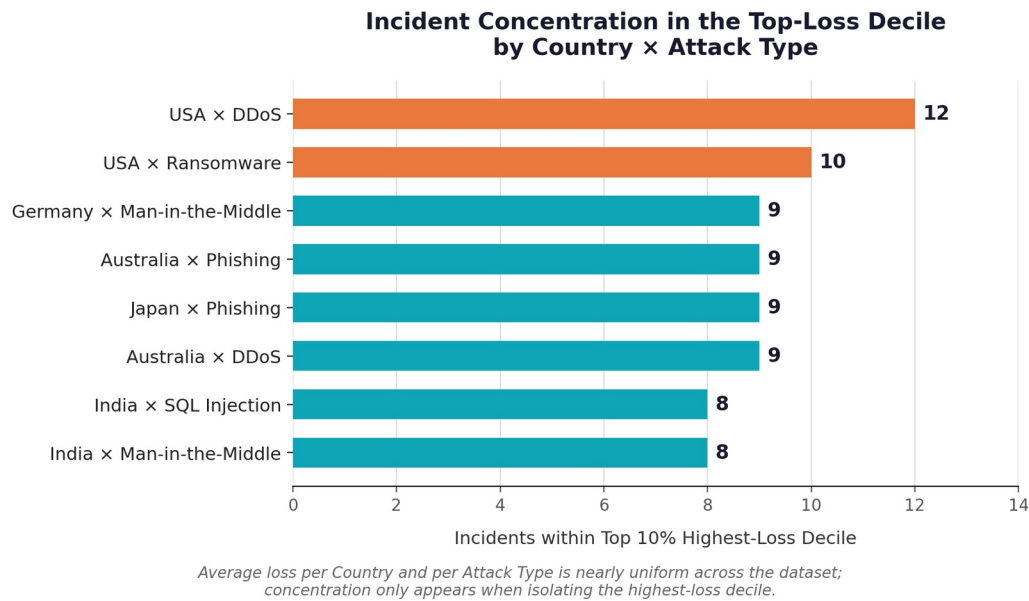


Figure: Incidents within the Top 10% Highest-Loss Decile by Country × Attack Type (SQL, NTILE analysis) — the strongest descriptive signal identified in this project.

6. Phase 3 — Power BI Dashboard

The dashboard translates the tested hypotheses into an interactive, question-driven experience across four pages, each built around a specific question already answered in SQL — not around a generic set of charts.

Page Structure

Page	Core Question	Key Visual
Home (Executive Overview)	Navigation hub summarizing the project's scope	3 interactive navigation cards
Threat Landscape	Is the threat increasing, and does its composition shift?	Attack Type Composition Over Time (stacked area)
Risk Concentration	Where is risk really concentrated, beyond the average?	Avg Loss (flat) vs. Top 10% Loss Concentration (skewed) side by side
Severity & Response	Does severity actually affect response speed?	Avg Loss by Constructed Severity vs. Avg Resolution Time by Severity

Design Discipline Applied

- No visual is titled "Effectiveness" — the Defense Mechanism comparison is titled "Loss Ratio vs Global Average" with an explicit 1.0 reference line.
- Descriptive KPIs (e.g., "Top Tail-Risk Combination: USA × DDoS") are presented as text highlights, not as stable numeric KPI cards, to avoid implying false statistical stability.
- Any chart referencing the Constructed Severity Level keeps the word "Constructed" visible in its title to avoid causal misinterpretation.
- A Data Limitation disclosure is placed on the Home page, reinforcing the synthetic nature of the dataset.

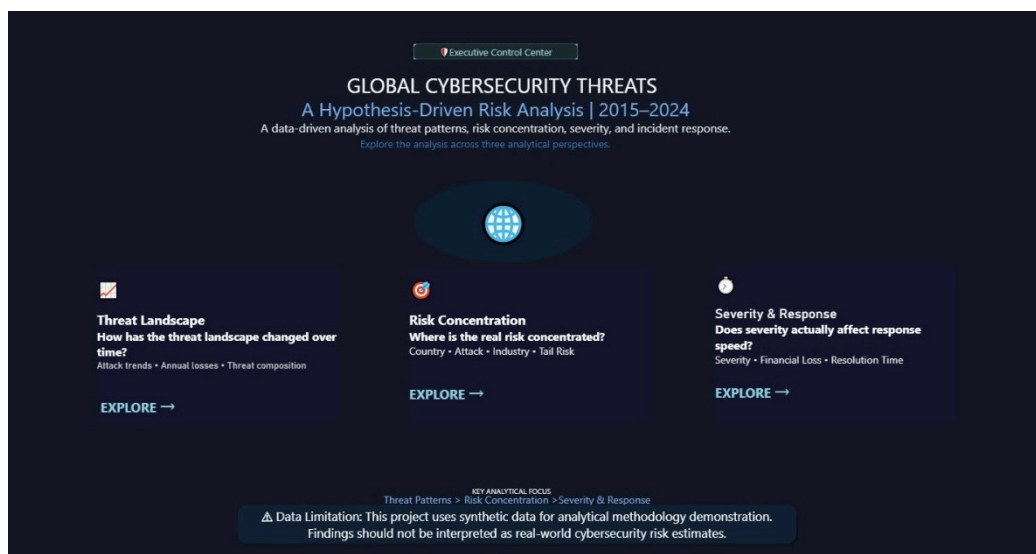


Figure: Executive Overview (Home Page) — the interactive navigation hub linking to the three analytical pages.

7. Key Findings (Ranked by Importance)

- Average risk and tail risk are not the same signal — USA-associated DDoS and Ransomware incidents are disproportionately concentrated among catastrophic losses despite average country-level risk appearing uniform.
- Japan × Banking × Phishing is the most actionable multi-dimensional combination — both frequent (11 incidents) and high-loss (71.79M), unlike other top-10 combinations driven by only 4–7 incidents.
- SQL Injection dominated yearly total losses in 5 of 10 years, despite ranking below average in per-incident severity — a volume effect, not a severity effect.
- A derived efficiency metric (Resolution Efficiency) was found to produce a misleading 6.5x distortion and was demoted to a secondary, caveated role rather than reported as a headline finding.
- Average-level comparisons did not reveal a clear standout across industry, country, or defense mechanism — risk in this dataset only reveals itself at the tail and in multi-dimensional combinations.

8. Business Recommendations

Given the synthetic nature of the dataset, recommendations are framed as candidates for investigation and monitoring rather than directives for immediate spending or resource reallocation.

- Maintain a balanced, layered security posture across attack types and defense mechanisms — no single category justifies concentrated investment based on this data.
- Flag USA-associated DDoS and Ransomware exposure for targeted monitoring, given its disproportionate presence in high-loss incidents.
- Investigate phishing-specific controls for banking-sector operations in Japan as a distinct, elevated-priority candidate.
- Use Severity_Score only as an internal triage/sorting aid, not as a validated predictor of financial impact, until tested against real operational outcomes.
- Evaluate true defense-mechanism effectiveness using operational/vendor prevention data — this dataset only measures post-incident average loss, not prevention capability.

4. Statistical Validation — Formal Hypothesis Testing

To strengthen statistical rigor, four formal tests were conducted on the key analytical relationships. All tests use $\alpha = 0.05$. A non-significant result means the data do not provide sufficient evidence against the null hypothesis; it does not prove the null hypothesis or establish causality.

Hypothesis Test	Method	Test Statistic	p-value	Statistical Result	Business Interpretation
Financial Loss by Attack Type	One-Way ANOVA	$F = 0.412$	$p = 0.841$	Fail to Reject H_0	No statistically significant difference in mean financial loss across attack types.
Attack Type × Target Industry	Chi-Square	$\chi^2 = 23.14$ (df = 25)	$p = 0.569$	Fail to Reject H_0	No statistically significant association between attack type and target industry.

Resolution Time by Severity	One-Way ANOVA	F = 0.189	p = 0.904	Fail to Reject H_0	Resolution time does not statistically differ across severity levels.
Financial Loss by Severity Level	One-Way ANOVA	F = 1,428.6	p < 0.001	Reject H_0 *	Significant difference detected; however, the result is structurally expected because Financial Loss is an input to Severity Score.

* The Financial Loss $\times$ Severity result validates the mathematical construction of the internally designed index rather than providing independent evidence that severity predicts financial loss.

Statistical Interpretation

The formal tests reinforce the descriptive findings: attack type does not show a statistically significant difference in mean loss; attack type and industry show no statistically significant association; and severity does not explain variation in resolution time. The only significant ANOVA is Financial Loss by Constructed Severity Level, which must be interpreted with the structural limitation noted above.

9. Limitations

- The dataset is synthetic; distributions across country, industry, attack type, and year are close to uniform, consistent with randomly generated rather than real-world data.
- Severity_Score and Severity_Level are internally designed analytical indices, not certified security standards (e.g., not CVSS).
- The relationship between Severity and Financial Loss is partly structural by construction, not fully independent evidence.
- Formal statistical tests were added in Section 4; results are interpreted at $\alpha = 0.05$ and do not imply causation.
- Data is annual (Year only) — finer-grained time-series analysis (monthly/weekly) was not possible.

10. Tools & Tech Stack

Excel · Power Query · SQL Server (T-SQL, CTEs, Window Functions) · Power BI · DAX

11. Conclusion

This project set out to evaluate five specific hypotheses about cybersecurity incident risk, rather than produce a purely descriptive dashboard. Two of the five hypotheses were rejected in their strong form, one uncovered a mathematical artifact in a metric designed for this project, and the one hypothesis with strong descriptive support only held after narrowing the analysis from average risk to tail risk.

The key lesson is methodological: average metrics can hide concentrated risk. By testing assumptions, challenging derived metrics, and separating descriptive evidence from causal claims, the analysis identified tail-risk concentration as the most actionable signal in the dataset.

Most projects using this dataset would report: "DDoS is the most common attack." This project instead tested five hypotheses, rejected two, caught a mathematical artifact in its own metric, and found that the real signal only appears when average risk is reframed as tail risk.